

Practice Quiz: The Invention as a Living System

Part A: Multiple Choice

1. In Active Inference, a Markov blanket separates an invention's internal states from its external states. Which of the following is an example of a **sensory state** in a household thermostat? A) The heating coil that warms the room B) The temperature sensor that measures room temperature C) The desired temperature set by the user D) The wiring connecting the thermostat to the furnace
2. Thomas Edison designed his light bulb with a high-resistance filament. Viewed from the Markov blanket of the bulb alone, this seems suboptimal (dimmer light). At what system level does this decision minimize free energy? A) The filament level — high resistance reduces filament wear B) The bulb level — high resistance produces better color temperature C) The distribution network level — high resistance allows thinner, cheaper copper wire D) The customer level — high resistance bulbs are easier to install
3. A smartphone's camera subsystem does not need to know the current cellular signal strength. This is an example of: A) Variational free energy minimization B) Conditional independence between nested subsystems C) Active state suppression D) Sensory boundary failure
4. An inventor designs a rain-sensing windshield wiper system. The system senses rain through an optical sensor on the windshield and acts by adjusting wiper speed. What is the perception-action loop? A) Rain falls -> sensor detects water -> controller adjusts speed -> wipers clear water -> sensor detects less water B) Driver presses button -> wipers activate -> rain stops -> driver releases button C) Rain falls -> wipers activate at maximum speed -> rain continues -> wipers stay at maximum D) Sensor calibrates itself -> wipers test their range -> system enters standby mode
5. The accuracy-complexity trade-off in invention design means: A) More accurate inventions are always more complex to build B) Complex inventions are always more accurate than simple ones C) An invention tuned precisely for one condition may fail in

others, while a simpler design may work adequately across many conditions D) Complexity should always be minimized regardless of accuracy

6. A bicycle's Markov blanket includes the handlebars and pedals as part of which category? A) Internal states — they are physical parts of the bicycle B) External states — they are touched by the rider, who is external C) Sensory and active boundary states — they transmit information and force between rider and bicycle mechanism D) They are not part of the Markov blanket because they are mechanical components
7. An inventor tries to create a universal kitchen tool that replaces knives, spatulas, whisks, tongs, and ladles. Based on the accuracy-complexity trade-off, what is the most likely outcome? A) The tool will excel at all five functions because more functions mean lower free energy B) The tool will perform each function adequately but none as well as dedicated tools, trading accuracy for breadth C) The tool will be simpler than any individual tool because it combines functions D) The tool will have a smaller Markov blanket because it replaces multiple tools

Part B: Short Answer and Analysis

1. Consider a simple invention: a doorbell. Map its Markov blanket by identifying (a) two internal states, (b) two external states, (c) one sensory state, and (d) one active state. Then describe the perception-action loop in one sentence.
2. A startup develops a smart plant pot that monitors soil moisture and sunlight, then sends notifications to the user's phone suggesting when to water or move the plant. However, the product receives complaints because the plant still dies — the system does not account for temperature, humidity, or the specific water needs of different plant species. Using the concept of system boundary placement, explain the failure and propose a specific boundary expansion (identify what new sensory states should be added and why).
3. Choose any invention from the past 50 years. Identify two nested subsystems within it and explain how their Markov blankets are related. Specifically: what information crosses the boundary between the two subsystems, and what information is kept conditionally independent?